

DAT NGUYEN

(408) 856-8287 | datq.nguyen06@gmail.com | linkedin.com/in/datnguy3n | github.com/Da0t | dats-nguyen.vercel.app

EDUCATION

University of California, San Diego

B.S. Data Science & B.A. Economics

La Jolla, CA

June 2028

Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Data Science, Data Management

Certifications: Operating Systems Principles, Docker & Kubernetes Principles, Intermediate SQL, Claude 101, AI Fluency

EXPERIENCE

Full-Stack Developer Intern

SEO (Sponsors for Educational Opportunity)

May 2026 – July 2026

New York, NY (Remote)

- Developed a full-stack Flask application in a SCRUM team that turns Zoom transcripts into assigned tasks through a human-in-the-loop OpenAI pipeline with schema validation, featuring role-based access control, Zoom and Google Calendar OAuth integrations, room-scoped real-time WebSocket updates, and Docker deployment, backed by 100+ pytest tests
- Engineered an installable Python CLI that debugs code by voice via Deepgram speech-to-text and the Claude API, with an automated mode that captures crash tracebacks from a sandboxed subprocess to diagnose and patch bugs, backed by a SQLite memory layer and a pytest-covered dependency-injected architecture
- Engineered a fault-tolerant reminder system with an idempotent database-deduplicated scheduled job for exactly-once delivery, with graceful degradation so third-party API failures never lose user data
- Completed an intensive full-stack engineering program with 35-40 hours/week of synchronous SWE training, weekly 1:1 engineering manager mentorship, and code reviews

Undergraduate Researcher

Economics Research Lab at UC San Diego

January 2026 – June 2026

La Jolla, CA

- Trained a Random Forest classifier to 0.982 ROC-AUC and 0.92 F1 on informal settlement detection, engineering geospatial features from 2,000+ Indonesia satellite images with QGIS, Python, and GeoPandas
- Extracted geospatial features from road networks, population density, and administrative boundaries across 239 balanced samples, reaching 96.6% recall and 87.8% precision on informal settlement detection

Software Engineer Lead

AI Student Collective at UC San Diego

September 2025 – May 2026

La Jolla, CA

- Developed and maintained a production platform serving 350+ members using Next.js, React, and Tailwind CSS, cutting load times 20% via ISR caching, streamlining deployments with CI/CD through GitHub Actions, and migrating event data from static JSON to Supabase PostgreSQL with schema design and RLS policies for non-technical club leads
- Improved PostgreSQL query performance 15% through execution plan analysis and column-level indexing on high-cardinality fields, and configured automated deploy testing via GitHub Actions to catch failed builds before release

PROJECTS

Aside AI | 1st Deepgram Track, Berkeley AI Hackathon | Python, Claude, Deepgram, Redis | GitHub

- Architected the backend for a real-time AI wearable, gating Claude vision behind a motion detector to cut API cost and latency, with a Redis-backed agent memory layer persisting narration history across a session
- Engineered a low-latency capture-to-narration pipeline spanning a QNX Raspberry Pi camera, a Python backend, and a React Native app over LAN, delivering spoken narration in 1–2 seconds

Pylon | 2nd Bow Capital Hackathon | Python, FastAPI, Next.js, TypeScript, deck.gl, Mapbox | Website

- Engineered a leaderless UDP gossip mesh in Python with NAT traversal, multi-hop flooding, and LRU deduplication, converging full network topology in under 2 seconds across nodes with no central server or single point of failure
- Built a terrain-aware simulation engine with BFS pathfinding, surface constraints, and gradient-based movement, surfaced through a real-time 3D tactical dashboard with 10+ custom deck.gl layers over Mapbox terrain

Rewind | 1st Place SDx Hackathon | Python, Next.js, TypeScript, Tailwind CSS, Omnara API | GitHub

- Created an agentic fork pipeline that paginates LLM message history, truncates at a user-chosen breakpoint, remaps schemas for Omnara's API, and resumes agent execution from the corrected state, with a typed 6-endpoint API client and regex-based fork detection requiring no extra LLM calls

TECHNICAL SKILLS

Languages: Python, Java, SQL, Bash, R, JavaScript/TypeScript, HTML/CSS

Frameworks & Libraries: React, Next.js, FastAPI, Tailwind CSS, Expo/React Native, PyTorch, Scikit-Learn, Pandas, SciPy, Matplotlib, Seaborn, statsmodels, Streamlit, deck.gl

Databases: PostgreSQL, SQLite, MySQL, Supabase, Firebase

Tools & Platforms: AWS (EC2, RDS, S3), Git/GitHub, Docker, Kubernetes, GitHub Actions, Databricks, Anaconda, Cloudflare, Nginx, Mapbox, QGIS, Claude Code, Codex, Redis